magnitude of diffusivities for gases, liquids, and solids.

2. Summarize the notation for the molecular, convective, and total fluxes for the three transport processes. How does one calculate the flux of mass, momentum, and energy across a surface with orientation **n**?

Mass:

$$|N\rangle = |J^*\rangle + C_\alpha|v^*\rangle \qquad \text{mole}$$

$$\int_V dV \frac{\partial}{\partial t}\rho_\alpha = -\int_S dS\, \rho_\alpha \langle v_\alpha|n\rangle + \int_V dV\, r_\alpha \qquad \text{mass}$$

accumulation

$$\int_V dV \frac{\partial C_\alpha}{\partial t} = -\int_S dS\, \overbrace{C_\alpha\langle v_\alpha|n\rangle}^{\text{combined mole flux}} + \int_V dV\, R_\alpha \qquad \text{mole}$$

Energy: (internal)

$$E = \rho u\langle v|n\rangle + \langle q|n\rangle$$

Momentum:

$$|\Phi| = \rho|v\rangle\langle v|n\rangle + |\pi|n\rangle$$

$$|\pi| = |\tau| + p|\delta|$$

Newton's law of viscosity: (constant ρ)

$$|\tau| = -\mu\left(|\nabla v| + |\nabla v|^t\right)$$

17B.2. Relations among fluxes in multicomponent systems. Verify Eqs. (K), (O), (T), and (X) of Table 17.8-1 using only the definitions of concentrations, velocities, and fluxes.

Table 17.8-1 Notation for Mass and Molar Fluxes*

Quantity	With respect to stationary axes		With respect to mass average velocity $\mathbf{v}$		With respect to molar average velocity $\mathbf{v}^*$	
Velocity of species α (cm/s)	$\mathbf{v}_\alpha$	(A)	$\mathbf{v}_\alpha - \mathbf{v}$	(B)	$\mathbf{v}_\alpha - \mathbf{v}^*$	(C)
Mass flux of species α (g/cm^2 s)	$\mathbf{n}_\alpha = \rho_\alpha \mathbf{v}_\alpha$	(D)	$\mathbf{j}_\alpha = \rho_\alpha(\mathbf{v}_\alpha - \mathbf{v})$	(E)	$\mathbf{j}_\alpha^* = \rho_\alpha(\mathbf{v}_\alpha - \mathbf{v}^*)$	(F)
Molar flux of species α (g-moles/cm^2 s)	$\mathbf{N}_\alpha = c_\alpha \mathbf{v}_\alpha$	(G)	$\mathbf{J}_\alpha = c_\alpha(\mathbf{v}_\alpha - \mathbf{v})$	(H)	$\mathbf{J}_\alpha^* = c_\alpha(\mathbf{v}_\alpha - \mathbf{v}^*)$	(I)
Sums of mass fluxes	$\sum_{\alpha=1}^{N} \mathbf{n}_\alpha = \rho \mathbf{v}$	(J)	$\sum_{\alpha=1}^{N} \mathbf{j}_\alpha = 0$	(K)	$\sum_{\alpha=1}^{N} \mathbf{j}_\alpha^* = \rho(\mathbf{v} - \mathbf{v}^*)$	(L)
Sums of molar fluxes	$\sum_{\alpha=1}^{N} \mathbf{N}_\alpha = c\mathbf{v}^*$	(M)	$\sum_{\alpha=1}^{N} \mathbf{J}_\alpha = c(\mathbf{v}^* - \mathbf{v})$	(N)	$\sum_{\alpha=1}^{N} \mathbf{J}_\alpha^* = 0$	(O)
Relations between mass and molar fluxes	$\mathbf{n}_\alpha = M_\alpha \mathbf{N}_\alpha$	(P)	$\mathbf{j}_\alpha = M_\alpha \mathbf{J}_\alpha$	(Q)	$\mathbf{j}_\alpha^* = M_\alpha \mathbf{J}_\alpha^*$	(R)
Interrelations among mass fluxes	$\mathbf{n}_\alpha = \mathbf{j}_\alpha + \rho_\alpha \mathbf{v}$	(S)	$\mathbf{j}_\alpha = \mathbf{n}_\alpha - \omega_\alpha \sum_{\beta=1}^{N} \mathbf{n}_\beta$	(T)	$\mathbf{j}_\alpha^* = \mathbf{n}_\alpha - x_\alpha \sum_{\beta=1}^{N} \frac{M_\alpha}{M_\beta} \mathbf{n}_\beta$	(U)
Interrelations among molar fluxes	$\mathbf{N}_\alpha = \mathbf{J}_\alpha^* + c_\alpha \mathbf{v}^*$	(V)	$\mathbf{J}_\alpha = \mathbf{N}_\alpha - \omega_\alpha \sum_{\beta=1}^{N} \frac{M_\beta}{M_\alpha} \mathbf{N}_\beta$	(W)	$\mathbf{J}_\alpha^* = \mathbf{N}_\alpha - x_\alpha \sum_{\beta=1}^{N} \mathbf{N}_\beta$	(X)

* Entries in the shaded boxes, involving the "hybrid fluxes" $\mathbf{j}_\alpha^*$ and $\mathbf{J}_\alpha$, are seldom needed; they are included only for the sake of completeness.

$$(K): \quad \sum_{\alpha=1}^{N} |j_\alpha\rangle = 0$$

$$\Rightarrow \quad j_\alpha = \rho_\alpha (|v_\alpha\rangle - |v\rangle)$$

$$\Rightarrow \quad \sum_{\alpha=1}^{N} |j_\alpha\rangle = \sum_{\alpha=1}^{N} \rho_\alpha (|v_\alpha\rangle - |v\rangle)$$

$$\Rightarrow \quad \sum_{\alpha=1}^{N} |j_\alpha\rangle = \sum_{\alpha=1}^{N} \underbrace{\rho_\alpha |v_\alpha\rangle}_{\rho|v\rangle} - |v\rangle \underbrace{\sum_{\alpha=1}^{N} \rho_\alpha}_{\rho}$$

$$\Rightarrow \quad \sum_{\alpha=1}^{N} |j_\alpha\rangle = \rho|v\rangle - |v\rangle \cdot \rho = 0$$

$$(O): \quad \sum_{\alpha=1}^{N} |J_\alpha^*\rangle = 0$$

$$\Rightarrow \quad |J_\alpha^*\rangle = c_\alpha (|v_\alpha\rangle - |v^*\rangle)$$

$$\Rightarrow \quad \sum_{\alpha=1}^{N} |J_\alpha^*\rangle = \sum_{\alpha=1}^{N} c_\alpha (|v_\alpha\rangle - |v^*\rangle)$$

$$\Rightarrow \quad \sum_{\alpha=1}^{N} |J_\alpha^*\rangle = \sum_{\alpha=1}^{N} \underbrace{c_\alpha |v_\alpha\rangle}_{c\cdot|v^*\rangle} - |v^*\rangle \underbrace{\sum_{\alpha=1}^{N} c_\alpha}_{c} = c|v^*\rangle - |v^*\rangle c = 0$$

(7): $\quad j_\alpha = |n_\alpha\rangle - w_\alpha \sum_{\beta=1}^{N} |n_\beta\rangle$

$j_\alpha = \rho_\alpha (|v_\alpha\rangle - |v\rangle)$, $|n_\alpha\rangle = \rho_\alpha |v_\alpha\rangle$, $\sum_{\beta=1}^{N} |n_\beta\rangle = \rho |v\rangle$

$w_\alpha = \frac{\rho_\alpha}{\rho}$

$\Rightarrow \rho_\alpha = w_\alpha \rho$

$\Rightarrow |j_\alpha\rangle = \rho_\alpha (|v_\alpha\rangle - |v\rangle)$

$|j_\alpha\rangle = \rho_\alpha |v_\alpha\rangle - \rho_\alpha |v\rangle$

$|j_\alpha\rangle = |n_\alpha\rangle - \rho_\alpha |v\rangle$

$|j_\alpha\rangle = |n_\alpha\rangle - w_\alpha \rho_\alpha |v\rangle$

$|j_\alpha\rangle = |n_\alpha\rangle - w_\alpha \sum_{\beta=1}^{N} |n_\beta\rangle$

(X): $\quad |J_\alpha^*\rangle = N_\alpha - x_\alpha \sum_{\beta=1}^{N} |N_\beta\rangle$

$J_\alpha^* = C_\alpha (|v_\alpha\rangle - |v^*\rangle)$, $|N_\alpha\rangle = C_\alpha |v_\alpha\rangle$, $x_\alpha = \frac{C_\alpha}{C}$, $\sum_{\beta=1}^{N} |N_\beta\rangle = C |V^*\rangle$

$\Rightarrow C_\alpha = x_\alpha C$

$\Rightarrow |J_\alpha^*\rangle = C_\alpha (|v_\alpha\rangle - |v^*\rangle)$

$\Rightarrow |J_\alpha^*\rangle = C_\alpha |v_\alpha\rangle - C_\alpha |v^*\rangle$

$|J_\alpha^*\rangle = |N_\alpha\rangle - C_\alpha |v^*\rangle$

$|J_\alpha^*\rangle = |N_\alpha\rangle - x_\alpha \underbrace{C |v^*\rangle}_{\Delta}$

$|J_\alpha^*\rangle = |N_\alpha\rangle - x_\alpha \sum_{\beta=1}^{N} |N_\beta\rangle$

17B.3. Relations between fluxes in binary systems. The following equation is useful for interrelating expressions in mass units and those in molar units in two-component systems:

$$\frac{\mathbf{j}_A}{\rho\omega_A\omega_B} = \frac{\mathbf{J}_A^*}{cx_Ax_B} \qquad (17B.3\text{-}1)$$

Verify the correctness of this relation.

Start w/:

$$j_\alpha = |n_A\rangle - \omega_\alpha(|n_A\rangle + |n_B\rangle)$$

$$\Rightarrow \quad j_\alpha = \omega_B |n_A\rangle - \omega_A |n_B\rangle$$

$$\Rightarrow \quad |n_A\rangle = (MW_A)|N_A\rangle \quad , |n_B\rangle = (MW_B)|N_B\rangle$$

$$(1) \quad |j_A\rangle = \omega_B (MW_A)|N_A\rangle - \omega_A(MW_B)|N_B\rangle$$

$$\underset{(x)}{\Rightarrow} \quad |J_A^*\rangle = |N_A\rangle - x_A(|N_A\rangle + |N_B\rangle)$$

$$|J_A^*\rangle = x_B|N_A\rangle - x_A \cdot |N_B\rangle$$

and $$\omega_A = \frac{(MW_A)C_A}{\rho} \quad , \quad \omega_B = \frac{(MW_B)C_B}{\rho}$$

$$x_A = \frac{C_A}{C} \quad , \quad x_B = \frac{C_B}{C}$$

$$(1) \underset{\div \rho\omega_A\omega_B}{\Rightarrow} \quad \frac{|j_A\rangle}{\rho\omega_A\omega_B} = \frac{\cancel{\omega_B}(MW_A)|N_A\rangle}{\rho\omega_A\cancel{\omega_B}} - \frac{\cancel{\omega_A}(MW_B)|N_B\rangle}{\rho\cancel{\omega_A}\omega_B}$$

$$\Rightarrow \quad \frac{|j_A\rangle}{\rho\omega_A\omega_B} = \frac{(\cancel{MW_A})|N_A\rangle}{\cancel{\rho}\frac{(\cancel{MW_A})\cdot C_A}{\cancel{\rho}}} - \frac{(\cancel{MW_B})|N_B\rangle}{\cancel{\rho}\cdot\frac{(\cancel{MW_B})C_B}{\cancel{\rho}}}$$

$$\Rightarrow \quad \boxed{\frac{|j_A\rangle}{\rho\omega_A\omega_B}} = \frac{|N_A\rangle}{x_A C} - \frac{|N_B\rangle}{x_B C} = \frac{\overbrace{x_B|N_A\rangle - x_A|N_B\rangle}^{|J_A^*\rangle}}{C x_A x_B} = \boxed{\frac{|J_A^*\rangle}{C x_A x_B}}$$

17B.4. Equivalence of various forms of Fick's law for binary mixtures.

(a) Starting with Eq. (A) of Table 17.8-2, derive Eqs. (B), (D), and (F).

(b) Starting with Eq. (A) of Table 17.8-2, derive the folowing flux expressions:

$$\mathbf{j}_A = -\rho(M_A M_B/M^2)\mathscr{D}_{AB}\nabla x_A \qquad (17B.4\text{-}1)$$

$$\nabla x_A = \frac{1}{c\mathscr{D}_{AB}}(x_A \mathbf{N}_B - x_B \mathbf{N}_A) \qquad (17B.4\text{-}2)$$

What conclusions can be drawn from these two equations?

(c) Show that Eq. (F) of Table 17.8-2 can be written as

$$\mathbf{v}_A - \mathbf{v}_B = -\mathscr{D}_{AB}\nabla \ln \frac{x_A}{x_B} \qquad (17B.4\text{-}3)$$

Table 17.8-2 Equivalent Forms of Fick's (First) Law of Binary Diffusion

Flux	Gradient	Form of Fick's Law	
$\mathbf{j}_A$	$\nabla\omega_A$	$\mathbf{j}_A = -\rho\mathscr{D}_{AB}\nabla\omega_A$	(A)
$\mathbf{J}_A^*$	∇x_A	$\mathbf{J}_A^* = -c\mathscr{D}_{AB}\nabla x_A$	(B)
$\mathbf{n}_A$	$\nabla\omega_A$	$\mathbf{n}_A = \omega_A(\mathbf{n}_A + \mathbf{n}_B) - \rho\mathscr{D}_{AB}\nabla\omega_A = \rho_A\mathbf{v} - \rho\mathscr{D}_{AB}\nabla\omega_A$	(C)
$\mathbf{N}_A$	∇x_A	$\mathbf{N}_A = x_A(\mathbf{N}_A + \mathbf{N}_B) - c\mathscr{D}_{AB}\nabla x_A = c_A\mathbf{v}^* - c\mathscr{D}_{AB}\nabla x_A$	(D)
$\rho(\mathbf{v}_A - \mathbf{v}_B)$	$\nabla\omega_A$	$\rho(\mathbf{v}_A - \mathbf{v}_B) = -\frac{\rho\mathscr{D}_{AB}}{\omega_A\omega_B}\nabla\omega_A$	(E)
$c(\mathbf{v}_A - \mathbf{v}_B)$	∇x_A	$c(\mathbf{v}_A - \mathbf{v}_B) = -\frac{c\mathscr{D}_{AB}}{x_A x_B}\nabla x_A$	(F)

a) $|j_A\rangle = -\rho D_{AB}|\nabla w_A\rangle \quad (A)$

* from 17.B3: $\dfrac{|j_A\rangle}{\rho w_A w_B} = \dfrac{|J_A^*\rangle}{c\, x_A x_B}$

$\Rightarrow |J_A^*\rangle = \dfrac{c x_A x_B}{\rho w_A \cdot w_B}|j_A\rangle$

sub (A) $\Rightarrow$ r $|j_A\rangle$

$|J_A^*\rangle = \dfrac{c x_A x_B}{\rho w_A w_B}(-\rho D_{AB}|\nabla w_A\rangle)$

$\Rightarrow |J_A^*\rangle = -c D_{AB}\left(\underbrace{\dfrac{x_A x_B}{w_A w_B}|\nabla w_A\rangle}\right)$

$\Rightarrow$ for binary systems:

$|J_A^*\rangle = -c D_{AB}|\nabla x_A\rangle$

(D): $|N_A\rangle = x_A(|N_A\rangle + |N_B\rangle) - c D_{AB}|\nabla x_A\rangle$

$\Rightarrow |J_A^*\rangle = |N_A\rangle - x_A(|N_A\rangle + |N_B\rangle)$

$|N_A\rangle = x_A(|N_A\rangle + |N_B\rangle) + |J_A^*\rangle$

(B) $\Rightarrow |N_A\rangle = x_A(|N_A\rangle + |N_B\rangle) - c D_{AB}|\nabla x_A\rangle$

(F): $c(|v_A\rangle - |v_B\rangle) = -\frac{c\rho_{AB}}{x_A x_B}|\nabla x_A\rangle$

$\Rightarrow |J_A^*\rangle = c_A(|v_A\rangle - |v^*\rangle)$

$|J_A^*\rangle = c x_A(|v_A\rangle - |v^*\rangle)$, $|v^*\rangle = x_A|v_A\rangle + x_B|v_B\rangle$

$\Rightarrow |v_A\rangle - |v^*\rangle = |v_A\rangle - (x_A|v_A\rangle + x_B|v_B\rangle)$

$|v_A\rangle - |v^*\rangle = (1-x_A)|v_A\rangle - x_B|v_B\rangle$

$1-x_A = x_B$
$\Rightarrow |v_A\rangle - |v^*\rangle = x_B|v_A\rangle - x_B|v_B\rangle$

sub $\Rightarrow |J_A^*\rangle = c x_A x_B(|v_A\rangle - |v_B\rangle)$

(B) $\Rightarrow c x_A x_B(|v_A\rangle - |v_B\rangle) = -c D_{AB}|\nabla x_A\rangle$

$c\cdot(|v_A\rangle - |v_B\rangle) = -\frac{c D_{AB}}{x_A x_B}|\nabla x_A\rangle$

6) $|j_A\rangle = -\rho D_{AB}|\nabla w_A\rangle$

for binary mix: $w_A = \frac{x_A M_A}{x_A M_A + x_B M_B} \Rightarrow |\nabla w_A\rangle = \frac{M_A M_B}{(x_A M_A + x_B M_B)^2}|\nabla x_A\rangle$

and $M_{TOT} \Rightarrow M = x_A M_A + x_B M_B$

$\Rightarrow |\nabla w_A\rangle = \frac{M_A M_B}{M^2}|\nabla x_A\rangle$

sub into (A) $\Rightarrow$ $j_A = -\rho D_{AB}\left(\frac{M_A M_B}{M^2}|\nabla x_A\rangle\right)$

$\Rightarrow \boxed{j_A = -\rho\left(\frac{M_A M_B}{M^2}\right) D_{AB}|\nabla x_A\rangle}$

$$|\nabla x_A\rangle = \frac{-M^2}{\rho M_A M_B D_{AB}} j^A$$ and $\rho = c \cdot M = \frac{mol}{L} \cdot \frac{g}{mol} = \frac{g}{L}$

$$|\nabla x_A\rangle = -\frac{M}{c M_A M_B D_{AB}} \cdot j^A$$, $w_A = \frac{x_A M_A}{M}$, $w_B = \frac{x_B M_B}{M}$

$$\Rightarrow \frac{M}{M_A M_B} = \frac{x_A}{M_B w_A} = \frac{x_B}{M_A w_B}$$

combine

$$\Rightarrow \quad |\nabla x_A\rangle = \frac{1}{c \cdot D_{AB}} \Big($$

c) (F) : $c(|v_A\rangle - |v_B\rangle) = -\frac{cD_{AB}}{x_A x_B} |\cdot \nabla x_A\rangle$

$\div c \implies$ $|v_A\rangle - |v_B\rangle = -\frac{D_{AB}}{x_A x_B} \cdot |\nabla x_A\rangle$, $x_A + x_B = 1$

$\Rightarrow \nabla x_B = -\nabla x_A$

note: $\frac{1}{x_A x_B}\nabla x_A = \frac{1}{x_A}\nabla x_A - \frac{1}{x_B}\nabla x_B$

$\Rightarrow$ $\ln x_A = \frac{1}{x_A}\nabla x_A$

$\ln x_B = \frac{1}{x_B}\nabla x_B$

$\Rightarrow \frac{1}{x_A x_B}\nabla x_A = \nabla \ln x_A - \nabla \ln x_B = \nabla \ln\left(\frac{x_A}{x_B}\right)$

sub $\implies$ $|v_A\rangle - |v_B\rangle = -D_{AB} \cdot \nabla \ln\left(\frac{x_A}{x_B}\right)$

18B.2. Error in neglecting the convection term in evaporation.

(a) Rework the problem in the text in §18.2 by neglecting the term $x_A(N_A + N_B)$ in Eq. 18.0-1. Show that this leads to

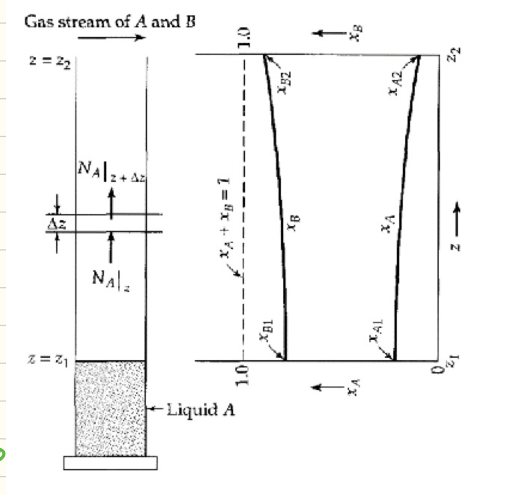
$$N_{Az} = \frac{c\mathscr{D}_{AB}}{z_2 - z_1}(x_{A1} - x_{A2}) \qquad (18B.2\text{-}1)$$

This is a useful approximation if A is present only in very low concentrations.

(b) Obtain the result in (a) from Eq. 18.2-14 by making the appropriate approximation.

(c) What error is made in the determination of $\mathscr{D}_{AB}$ in Example 18.2-1 if the result in (a) is used?

Answer: 5%



a) 18.2: $N_{Az} = -\frac{c D_{AB}}{1-x_A}\frac{dx_A}{dz}$

Eq: (18.0-1) $\underbrace{N_{Az}}_{\text{combined flux}} = \underbrace{-cD_{AB}\frac{\partial x_A}{\partial z}}_{\text{Diffusive flux}} + \underbrace{\cancel{x_A(N_{Az}+N_{Bz})}}_{\text{Convective flux}}$

Assumptions:
- S.S
- No B motion

* Neglecting convection:

$N_{Az} = -cD_{AB}\cdot\frac{dx_A}{dz}$ & $\frac{d^2x_A}{dz^2} = 0$

for ideal gas mole fraction gradient wrt z direction is constant

$\frac{d^2x_A}{dz^2} = 0 \Rightarrow x_A = C_1 z + C_2$

BCs: $z = z_1 \quad x_A = x_{A1}, \; x_B = x_{B1}$

$z = z_2 \quad x_A = x_{A2}, \; x_B = x_{B2}$

apply BCs solve for C_1, C_2:

$x_{A1} = C_1 z_1 + C_2$

$x_{A2} = C_1 z_2 + C_2$

$\Rightarrow C_1 = \frac{x_{A2}-x_{A1}}{z_2 - z_1}$

$C_2 = x_{A1} - \left(\frac{x_{A2}-x_{A1}}{z_2-z_1}\right)z_1$

$$\Rightarrow x_A(z) = \left(\frac{x_{A2}-x_{A1}}{z_2-z_1}\right)z + x_{A1} - \left(\frac{x_{A2}-x_{A1}}{z_2-z_1}\right)z_1$$

$$\Rightarrow \boxed{\frac{x_A - x_{A1}}{x_{A2}-x_{A1}} = \frac{z - z_1}{z_2 - z_1}}$$

b) 18.2-14: $N_{Az}\big|_{z=z_1} = \dfrac{-cD_{AB}}{1-x_{A1}} \dfrac{dx_A}{dz}\Big|_{z=z_1} = +\dfrac{cD_{AB}}{x_B}\dfrac{dx_B}{dz}\Big|_{z=z_1}$

$$= \frac{cD_{AB}}{z_2-z_1}\ln\left(\frac{x_{B2}}{x_{B1}}\right)$$

$$\Rightarrow \quad N_{Az}\big|_{z=z_1} = \frac{cD_{AB}}{(z_2-z_1)(x_B)_{ln}}(x_{A1}-x_{A2})$$

$$\Rightarrow 18.2\text{-}16 \quad N_{Az}\big|_{z=z_1} = \frac{cD_{AB}(x_{A1}-x_{A2})}{(z_2-z_1)}\left[1+\frac{1}{2}(x_{A1}+x_{A2})+\frac{1}{3}(x_{A1}^2+x_{A1}x_{A2}+x_{A2}^2)\ldots\right]$$

$\Rightarrow$ if x_{A1} is very low at $z=z_1$

$$\Rightarrow \quad N_{Az}\big|_{z=z_1} = -cD_{AB}\left(\frac{x_{A2}-x_{A1}}{z_2-z_1}\right) \approx -cD_{AB}\cdot\frac{dx_A}{dz}$$

c) for ideal mixture:

van't Hoff eqn: $p = cRT$

$$D_{AB} = -\frac{N_{Az}}{c}\cdot\frac{dz}{dx_A} = -\left(\frac{N_{Az}RT}{p}\right)\left(\frac{z_2-z_1}{x_{A2}-x_{A1}}\right)$$

from 18.2-2:

$\Delta z = 17.1$ cm

$x_{A1} = \dfrac{33}{755}$

$p_{TOT} = 755$ mmHg $= 755/760$ atm

$T = 0°C = 273.15$ K

$R = 82.06$ cm^3 atm/mol·K

$N_A = 7.26\times10^{-9}$ gmole/cm^2·s

18.2-2: $D_{AB} = 0.0636$ cm^2/s

$$D_{AB} = -\left(\frac{N_{Az}RT}{p}\right)\left(\frac{z_2-z_1}{x_{A2}-x_{A1}}\right)$$

$$= -\frac{(7.26\times10^{-9})(82.06)(273.15)}{(755/760)}\left(\frac{17.1}{0-\left(\frac{33}{755}\right)}\right)$$

$= 0.06409$ gmol/cm^2·s

err% $= \dfrac{0.06409-0.0636}{0.0636}\times100 = 0.77\%$

18B.5. Absorption of chlorine by cyclohexene. Chlorine can be absorbed from Cl_2–air mixtures by olefins dissolved in CCl_4. It was found[5] that the reaction of Cl_2 with cyclohexene (C_6H_{10}) is second order with respect to Cl_2 and zero order with respect to C_6H_{10}. Hence the rate of disappearance of Cl_2 per unit volume is $k_2'''c_A^2$ (where A designates Cl_2).

Rework the problem of §18.4 where B is a C_6H_{10}–CCl_4 mixture, assuming that the diffusion can be treated as pseudobinary. Assume that the air is essentially insoluble in the C_6H_{10}–CCl_4 mixture. Let the liquid phase be sufficiently deep that L can be taken to be infinite.

(a) Show that the concentration profile is given by

$$\frac{c_{A0}}{c_A} = \left[1 + \sqrt{\frac{k_2'''c_{A0}}{6\mathscr{D}_{AB}}}\,z\right]^2 \qquad (18B.5\text{-}1)$$

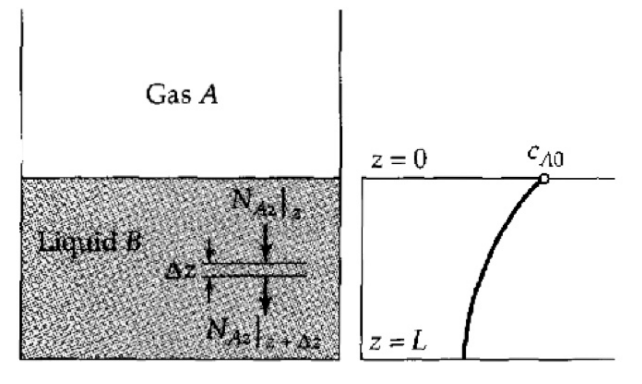
(b) Obtain an expression for the rate of absorption of Cl_2 by the liquid.

(c) Suppose that a substance A dissolves in and reacts with substance B so that the rate of disappearance of A per unit volume is some arbitrary function of the concentration, $f(c_A)$. Show that the rate of absorption of A is given by

$$N_{Az}|_{z=0} = \sqrt{2\mathscr{D}_{AB}\int_0^{c_{A0}} f(c_A)\,dc_A} \qquad (18B.5\text{-}2)$$

Use this result to check the result of (b).

[5] G. H. Roper, *Chem. Eng. Sci.*, **2**, 18–31, 247–253 (1953).



a) MB: $S\cdot N_{Az}\big|_z - S\cdot N_{Az}\big|_{z+\Delta z} + \overbrace{S\cdot\Delta z}^{\text{volume}} \cdot R_A = 0$

and $N_{Az} = \langle J_A^* \rangle = -D_{AB}\dfrac{dC_A}{dz}$, $R_A = \dfrac{dC_A}{dt} = -k_2'' C_A^2$

÷ volume: $S\cdot\Delta z$

$\Rightarrow$

$$\lim_{\Delta z\to 0}\left(-\left[\frac{N_{Az}|_{z+\Delta z} - N_{Az}|_z}{\Delta z}\right] + k_2''' C_A^2\right) = 0$$

$$-\lim_{\Delta z\to 0}\left(\frac{N_{Az}|_{z+\Delta z} - N_{Az}|_z}{\Delta z} + k_2''' C_A^2\right) = 0$$

$$\Rightarrow \frac{dN_{Az}}{dz} + k_2'' C_A^2 = \frac{d}{dz}\left(-D_{AB}\frac{dC_A}{dz}\right) + k_2''' C_A^2 = 0$$

BCs: @ $z=0$ $C_A = C_{A0}$

@ $z = L$ $N_{Az} = 0$ $\left(\dfrac{dC_A}{dz} = 0\right)$

$\Rightarrow$ Making dimensionless:

$\Gamma = \dfrac{C_A}{C_{A0}}$, $\zeta = \dfrac{z}{L}$, thiele modulus: $\phi = \left(\dfrac{k_2''' \cdot L^2}{D_{AB}}\right)^{1/2}$

$$\Rightarrow \quad \left(\frac{C_{A0}}{L^2}\right)\frac{d^2\Gamma}{d\zeta^2} - \left(\frac{k_2''' \cdot C_{A0}^2}{D_{AB}}\right)\Gamma^2 = 0$$

$$\xrightarrow[\text{sub } \phi]{\cdot\left(\frac{C_{A0}}{L}\right)} \quad \frac{d^2\Gamma}{d\zeta^2} - C_{A0}\cdot\phi^2\cdot\Gamma^2 = 0$$

New BCs:

@ $\zeta = 0$ $(z=0)$ $\Gamma = 1$ $(C_A = C_{A0})$

@ $\zeta = 1$ $(z=L)$ $\frac{d\Gamma}{d\zeta} = 0$ $(N_{Az} = 0)$

$\Rightarrow$ Separation of variables:

$$f = \frac{d\Gamma}{d\zeta} \Rightarrow \frac{d^2\Gamma}{d\zeta^2} = \frac{df}{d\zeta} = \frac{df}{d\Gamma}\cdot\left(\frac{d\Gamma}{d\zeta}\right) = f\cdot\frac{df}{d\Gamma}$$

$$\Rightarrow \quad f\cdot\frac{df}{d\Gamma} - C_{A0}\phi^2\cdot\Gamma^2 = 0$$

$$\Rightarrow \quad \int_0^f f\,df = C_A\phi^2\int_0^\Gamma \Gamma^2\,d\Gamma$$

$$\Rightarrow \quad \frac{1}{2}f^2 = \frac{1}{3}C_{A0}\phi^2\cdot\Gamma^3$$

$$\Rightarrow \quad f = \frac{d\Gamma}{d\zeta} = \pm\sqrt{\frac{2}{3}C_{A0}\phi^2}\cdot\Gamma^{3/2}$$

$\Rightarrow$ Since $\frac{dC_A}{dz}$ decreasing w/ z: $f = \frac{d\Gamma}{d\zeta} = -\sqrt{\frac{2}{3}C_{A0}\phi^2}\cdot\Gamma^{3/2}$

$$\Rightarrow \quad \int_1^\Gamma \Gamma^{-3/2}\,d\Gamma = -\sqrt{\frac{2}{3}C_A\phi^2}\cdot\int_0^\zeta d\zeta$$

$$\Rightarrow \quad -2\left(\Gamma^{-1/2} - 1\right) = -\sqrt{\frac{2}{3}C_{A0}\phi^2}\cdot\zeta$$

$$\Rightarrow \quad \frac{1}{\Gamma} = \left[1 + \sqrt{\frac{1}{6}C_{A0}\phi^2}\cdot\zeta\right]^2 \Rightarrow \frac{C_{A0}}{C_A} = \left[1 + \sqrt{\frac{k_2'''C_{A0}}{6\,D_{AB}}}\cdot z\right]^2$$

b) Rate of absorption of Cl_2:

$$N_{Az}\Big|_{z=0} = -\mathcal{D}_{AB}\frac{dC_A}{dz}\Big|_{z=0} = -\left(\frac{C_{A_0}\mathcal{D}_{AB}}{L}\right)\frac{d\Gamma}{d\zeta}\Big|_{\zeta=0} = \left(\frac{C_{A0}\mathcal{D}_{AB}}{L}\right)\sqrt{\frac{2}{3}C_{A0}\phi^2\cdot\Gamma^{3/2}}\Big|_{\Gamma=1}$$

$$N_{Az}\Big|_{z=0} = \sqrt{\frac{2}{3}k_2'''C_{A_0}^3\mathcal{D}_{AB}}$$

c)

$$-\mathcal{D}_{AB}\frac{\partial^2 C_A}{dz^2} + f(C_A) = 0 \quad \text{, let } p = \frac{dC_A}{dz}$$

$$\Rightarrow \frac{d^2 C_A}{dz} = \frac{d}{dz}\left(\frac{dC_A}{dz}\right) = \frac{dp}{dz} = \frac{dp}{dC_A}\cdot\left(\frac{dC_A}{dz}\right) = p\cdot\frac{dp}{dC_A}$$

$$\rightarrow \quad p\frac{dp}{dC_A} = \frac{f(C_A)}{\mathcal{D}_{AB}}$$

$$\Rightarrow \int_0^p p\,dp = \frac{1}{\mathcal{D}_{AB}}\int_0^{C_A} f(C_A)\,dC_A$$

$$\Rightarrow p = \frac{dC_A}{dz} = -\left[\frac{2}{\mathcal{D}_{AB}}\int_0^{C_A} f(C_A)\,dC_A\right]^{1/2}$$

(-) because $\frac{dC_A}{dz}$ decreasing wrt z

So $$N_{Az}\Big|_{z=0} = -\mathcal{D}_{AB}\frac{dC_A}{dz}\Big|_{z=0} = -\mathcal{D}_{AB}\left(\frac{-2}{\mathcal{D}_{AB}}\int_0^{C_A} f(C_A)\,dC_A\right)^{1/2}$$

$$N_{Az}\Big|_{z=0} = +\sqrt{2\mathcal{D}_{AB}\int_0^{C_A} f(C_A)\,dC_A}$$

$\rightarrow$ for $f(C_A) = k_2'''C_A^2 \Rightarrow \int_0^{C_{A0}} f(C_A)\,dC_A = k_2'''C_A^2\,dC_A = \frac{k_2'''}{3}\cdot C_{A0}^3$

$$\Rightarrow N_{Az}\Big|_{z=0} = \sqrt{2\mathcal{D}_{AB}\int_0^{C_{A0}} f(C_A)\,dC_A} = \sqrt{\frac{2}{3}k_2'''C_{A0}^3\mathcal{D}_{AB}}$$

18B.8. Method for separating helium from natural gas (Fig. 18B.8). Pyrex glass is almost impermeable to all gases but helium. For example, the diffusivity of He through pyrex is about 25 times the diffusivity of H_2 through pyrex, hydrogen being the closest "competitor" in the diffusion process. This fact suggests that a method for separating helium from natural gas could be based on the relative diffusion rates through pyrex.[7]

Suppose a natural gas mixture is contained in a pyrex tube with dimensions shown in the figure. Obtain an expression for the rate at which helium will "leak" out of the tube, in terms the diffusivity of helium through pyrex, the interfacial concentrations of the helium in the pyrex, and the dimensions of the tube.

Answer: $W_{He} = 2\pi L \dfrac{\mathscr{D}_{He\text{-}Pyrex}(c_{He,1} - c_{He,2})}{\ln(R_2/R_1)}$

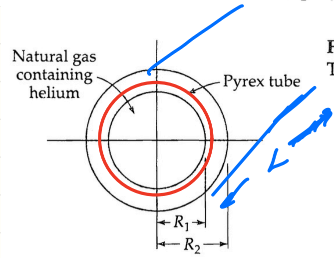


Fig. 18B.8. Diffusion of helium through pyrex tubing. The length of the tubing is L.

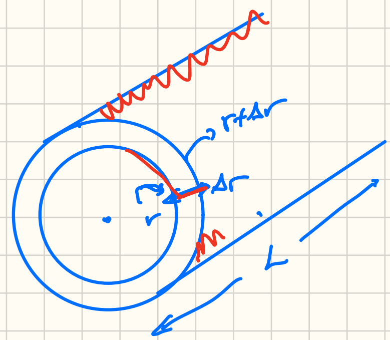
Assumptions:

- N_{Ar}, no N_{Az}, $N_{A\theta}$
- $x_A = x_{He} \ll 1$
- $N_{Bz} = 0$
- Convection is neglected
- $C_A(r)$

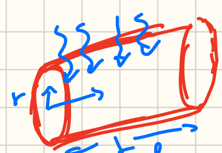
He : A

Pyrex : B

$\Big\}\ D_{He\text{-}Pyrex} = D_{AB}$



volume : $2\pi r \Delta r L$



area of diffusion ⊥ r

so base : $2\pi r \cdot L$
(circumference) length along tube

Shell balance:

$$\frac{\cancel{2\pi r L}\cdot N_{Ar}\big|_r}{\cancel{2\pi r L}\,\Delta r} - \frac{\cancel{2\pi r L}\cdot N_{Ar}\big|_{r+\Delta r}}{\cancel{2\pi r L}\,\Delta r} \overset{S.S}{=} 0$$

$\div\, 2\pi r L \Delta r$

$\Rightarrow \lim_{\Delta r \to 0}$

$$\lim_{\Delta r \to 0} \frac{r N_{Ar}\big|_r - r N_{Ar}\big|_{r+\Delta r}}{\Delta r} = 0$$

$$= -\lim_{\Delta r \to 0} \frac{r N_{Ar}\big|_{r+\Delta r} + r N_{Ar}\big|_r}{\Delta r} = 0$$

$$= -\frac{d(rN_{Ar})}{dr} = 0$$

$$, \ N_A = C_A v^* + J_A^*$$

$$= C_A \sum x_\alpha v_\alpha + J_A^*$$

$$= \underbrace{\cancel{x_A N_A}}_{\ll 1} - D_{AB} \nabla C_A$$

$$\boxed{N_A = -D_{AB}\frac{dC_A}{dr}}$$

$$= \frac{d(rN_{Ar})}{dr} = 0$$

$$= d\left(r \cdot - D_{AB} \frac{dC_A}{dr}\right) = 0$$

B.C: $r = R_1 \quad C_{A1}$

$r = R_2 \quad C_{A2}$

$$= - D_{AB} \frac{d}{dr}\left(r \frac{dC_A}{dr}\right) = 0$$

$$= \int \frac{d}{dr}\left(r \frac{dC_A}{dr}\right) = \int 0$$

$$r \frac{dC_A}{dr} = C_1$$

$$\int \frac{dC_A}{dr} = \int \frac{C_1}{r}$$

$$C_A = C_1 \ln r + C_2$$

use BCs: $C_A(R_1) = C_1 \ln(R_1) + C_2$

$$C_{A1} = C_1 \ln(R_1) + C_2$$

$$C_{A2} = C_1 \ln(R_2) + C_2$$

$$\Rightarrow C_1 = \frac{C_{A1} - C_{A2}}{\ln\left(\frac{R_1}{R_2}\right)} , \quad C_2 = C_{A2} - (C_{A1} - C_{A2}) \frac{\ln R_2}{\ln\left(\frac{R_1}{R_2}\right)}$$

Conc profile for C_A:

$$\frac{C_A - C_{A2}}{C_{A1} - C_{A2}} = \frac{\ln\left(\frac{r}{R_2}\right)}{\ln\left(\frac{R_1}{R_2}\right)} = \frac{\ln\left(\frac{R_2}{r}\right)}{\ln\left(\frac{R_2}{R_1}\right)}$$

* Molar flow rate of helium (A) at interface $r = R$

multiply flux (per unit area/time) time Area $\Rightarrow$

molar flow rate (per time) * at interface

Area is constant ($2\pi RL$) otherwise wed have to integrate:

$$W_A = (2\pi RL) \cdot N_{Ar}\Big|_{r=R} \quad \text{and} \quad N_{Ar} = -D_{AB}\frac{dC_A}{dr} = -D_{AB}\frac{C_1}{r}$$

$$N_{Ar} = \frac{D_{AB}}{r}\frac{(C_{A1}-C_{A2})}{\ln(R_2/R_1)}.$$

$$\Rightarrow W_A = (2\pi \cancel{R} L) \cdot \frac{D_{AB}}{\cancel{R}} \cdot \frac{(C_{A1}-C_{A2})}{\ln(R_2/R_1)}.$$

$$\Rightarrow \boxed{W_A = 2\pi L \frac{D_{AB}(C_{A1}-C_{A2})}{\ln(R_2/R_1)}}$$